Literature Status: The Paulsen Problem

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11 August 2026

Status: literature already answered (central quantitative problem). Problem 1.1 of arXiv:1102.2344 asks for a quantitative perturbation from a nearly equal-norm, nearly Parseval frame to an equal-norm Parseval frame. Hamilton and Moitra explicitly identify this as the Paulsen problem, cite Cahill–Casazza, and prove the dimension-only bound $\mathrm{dist}^2 \leq 20\varepsilon d^2$ in arXiv:1809.04726.

1 Original question

Jameson Cahill and Peter G. Casazza, *The Paulsen Problem in Operator Theory*, arXiv:1102.2344, published in *Operators and Matrices* **7** (2013), 117–130, poses Problem 1.1 on arXiv PDF page 1 [1].

Source Problem 1.1. *Find a function $h(\varepsilon, M, N)$ such that every ε-nearly equal-norm, ε-nearly Parseval frame $\{f_i\}_{i=1}^N$ in an M-dimensional Hilbert space admits an equal-norm Parseval frame $\{g_i\}_{i=1}^N$ satisfying*

$$\sum_{i=1}^{N} \|f_i - g_i\|^2 \leq h(\varepsilon, M, N).$$

The displayed source has a harmless index typo, printing M rather than N on the family $\{g_i\}$; the sum and the frame cardinality make the intended indexing clear. Immediately after the problem, the authors call it fundamental to decide whether h can be independent of N.

2 Separate later answer

Linus Hamilton and Ankur Moitra, *The Paulsen Problem Made Simple*, arXiv:1809.04726, later published in *Israel Journal of Mathematics* **246** (2021), 299–313, explicitly states in its abstract and introduction that the Paulsen problem was resolved by Kwok–Lau–Lee–Ramachandran [2], and gives a simpler proof with a stronger estimate [3]. Its bibliography cites the Cahill–Casazza source paper by title.

On arXiv PDF page 3, Theorem 1 states:

Hamilton–Moitra, Theorem 1. *For every ε-nearly equal-norm Parseval frame V in $\mathbb{R}^d$, there is an equal-norm Parseval frame W such that*

$$\mathrm{dist}^2(V, W) = \sum_i \|v_i - w_i\|_2^2 \leq 20\varepsilon d^2.$$

Taking $d = M$ gives an admissible Paulsen function independent of the number N of frame vectors. Thus it resolves the source's central quantitative existence and N-independence question. The relation is author-explicit, not an agent-inferred reformulation.

Scope and search status

This packet records the standard real finite-dimensional formulation proved in the supporting paper. It does not claim that the bound $20\varepsilon d^2$ has optimal dependence on d: Hamilton–Moitra record a lower bound of order εd and leave closing that gap open. It also does not identify an answer to the prescribed-norm Generalized Paulsen Problem 5.3, Generalized Projection Problem 5.4, or the prescribed frame operator Problem 5.7 in the source paper.

A bounded search on 11 August 2026 checked the run registry, the local arXiv sources, the exact title and problem wording, and later Paulsen-problem papers. The decisive match is Hamilton–Moitra's arXiv:1809.04726, whose abstract says the problem was resolved and whose Theorem 1 supplies the explicit bound. The earlier resolving paper arXiv:1710.02587 gives $O(\varepsilon d^{13/2})$. No new mathematical result is claimed here.

References

[1] J. Cahill and P. G. Casazza, *The Paulsen Problem in Operator Theory*, Operators and Matrices **7** (2013), 117–130; arXiv:1102.2344.

[2] T. C. Kwok, L. C. Lau, Y. T. Lee, and A. Ramachandran, *The Paulsen Problem, Continuous Operator Scaling, and Smoothed Analysis*, STOC 2018, 182–189; arXiv:1710.02587.

[3] L. Hamilton and A. Moitra, *The Paulsen Problem Made Simple*, Israel J. Math. **246** (2021), 299–313; arXiv:1809.04726.